

FINAL EXAM PROJECT

Data Warehouse & Data Mining with Dashboard

Learning Objectives

By completing this project, students will be able to:

1. Prepare and integrate data into a data warehouse (ETL process).
2. Design a data warehouse schema (fact & dimension tables).
3. Perform data preprocessing for data mining.
4. Apply data mining techniques to extract insights.
5. Build an interactive dashboard for decision-making.

Project Scenario (Choose ONE)

1. Retail Sales Analysis
2. E-commerce Customer Behavior
3. Student Academic Performance
4. Hospital Management Data
5. Banking Transactions

Phase 1: Data Preparation (25%)

Tasks:

- Collect dataset
- Identify data sources, types, missing values
- Perform data cleaning, transformation, integration

Output:

- Clean dataset
- Data description document

Phase 2: Data Warehouse Design (25%)

Tasks:

- Define business requirements
- Design schema (Star Schema preferred)
- Create fact and dimension tables
- Build ERD/schema diagram
- Implement using SQL (MySQL/PostgreSQL/SQL Server)

Output:

- Data warehouse schema
- SQL scripts

Phase 3: Data Mining (25%)

Apply at least 1 techniques:

- Association Rule Mining
- Classification
- Clustering
- Prediction/Regression

Output:

- Model explanation
- Results and interpretation

Phase 4: Dashboard Development (15%)

Tools: Power BI, Tableau, Looker Studio, or Python (If Applicable) or Just a Mock Up of dashboard

Requirements:

- KPI summary
- At least 3 visualizations
- Filters
- Insight section

Output:

- Interactive dashboard

Phase 5: Report & Presentation (10%)

Report must include:

- Introduction
- Dataset description
- Data warehouse design
- Data mining process
- Dashboard explanation
- Business insights

Presentation: 10–15 minutes per group